

For NPRE 297, your total grade will be earned through multiple computational exercises as well as contribution to software projects, focused on advanced reactor modeling and simulation. This work will include data collection, collaboration, and potential code contributions to tools in the ARFC ecosystem. **For 1 credit hour, the student should contribute a minimum of 4 hours of effort per week.**

This independent study is intended to provide an informative and hands-on experience with data science, reproducible scientific computing, and software development for nuclear engineering purposes. The primary role will be to work in a small team to coordinate and collaborate on the Python packages MoSDeN and Ghastly. In MoSDeN, work will primarily consist of adding unit tests, refactoring functions and classes, and using Sphinx to automatically generate documentation. In Ghastly, work will primarily consist of beta testing and aiding with dataset version control. For both MoSDeN and Ghastly, the student will also be expected to also assist with reviewing Pull Requests (PRs) and creating issues. By the end of this independent study, the student will be expected to have gained experience with Python package development, version control with Git, and software testing and documentation. Data and software contributions resulting from this work will be contributed to the open source repositories at github.com/arfc. Deliverables will include: PRs the student created, PRs the student reviewed, issues the student created, and issues the student closed.

The student will be expected to consistently engage with the lead graduate student in this work as well as Prof. Huff. The deliverables will include commits to online code repositories as well as a final report and presentation summarizing accomplishments by the end of the semester. The project will be assessed as independent research work, much like a journal article undergoes peer review. I will be looking for :

- Relevance
- Technical Detail
- Analytic Rigor
- Verifiability
- Clarity
- A Conclusion

This work will consist of four deliverables:

- a work plan,
- completion of the Software Carpentry Git and GitHub training,
- collaborative code development toward software development in MoSDeN and Ghastly,
- a brief summary report in the style of a conference or journal paper,
- and a 10 minute summary presentation.

1. (10 points) **Work Plan: Due 2026.03.05**

To help establish scope and milestones, the first step of the project will be a work plan. This plan, developed under supervision of an Advanced Reactors and Fuel Cycles Research Group Graduate Research Assistant, will be delivered in the form of multiple GitHub issues with descriptions and deadlines describing the work you intend to do in the ARFC group. Start with completing the ARFC new member checklist.

Please arrange at least one short meeting with Prof. Huff to discuss the project plan so that I can help you refine your schedule. I would be happy to provide feedback on a draft of your plan (once) before it is due.

2. (10 points) **Git and GitHub Self-Study: Due 2026.03.19** Conduct the full novice tutorial hosted at Software Carpentry.

3. (60 points) **Software Contributions: Due 2026.05.06** Complete the full suite of GitHub Issues agreed upon as a part of the Work Plan, within the due dates specified in each issue. All issues must be completed by 2026.05.06. This will require responsive completion of all necessary pull requests in collaboration with colleagues within the ARFC group.

4. (20 points) **Final Report and Presentation: Due 2026.05.06** The final report must be submitted to Prof. Huff and a summary presentation will be given to the ARFC group during a weekly group meeting late in the semester.